PSEUDO-COHERENCE OF THE COMPLEXES IN A DISTINGUISHED TRIANGLE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lifting a map between complexes out of a bounded above complex of finite free modules. Derived Categories, Lemma \ref{derived-lemma-lift-map}, cited at proof lines 15934-15935. Produces the map between the two approximating complexes that makes the square commute in the derived category.
- The four and five lemmas for the comparison of long exact cohomology sequences. Homology, Lemma \ref{homology-lemma-four-lemma} and Homology, Lemma \ref{homology-lemma-five-lemma}, cited at proof lines 15947-15948. Convert the comparison map of triangles into the isomorphism and surjectivity statements on cohomology.

Source reference. Original lemma 064R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7875–7875. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 064E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7863–7863. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05QA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7369–7369. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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Source reference. Original lemma 05QB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7370–7370. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring and $m \in \mathbf{Z}$. Let $(K^\bullet, L^\bullet, M^\bullet, f, g, h)$ be a distinguished triangle in $D(R)$.*

1. *If $K^\bullet$ is $(m+1)$-pseudo-coherent and $L^\bullet$ is m-pseudo-coherent then $M^\bullet$ is m-pseudo-coherent.*
2. *If $K^\bullet, M^\bullet$ are m-pseudo-coherent, then $L^\bullet$ is m-pseudo-coherent.*
3. *If $L^\bullet$ is $(m+1)$-pseudo-coherent and $M^\bullet$ is m-pseudo-coherent, then $K^\bullet$ is $(m+1)$-pseudo-coherent.*

Proof. Proof of (1). Choose $\alpha : P^\bullet \to K^\bullet$ with $P^\bullet$ a bounded complex of finite free modules such that $H^i(\alpha)$ is an isomorphism for $i > m+1$ and surjective for $i = m+1$. We may replace $P^\bullet$ by $\sigma_{\geq m+1} P^\bullet$ and hence we may assume that $P^i = 0$ for $i < m+1$. Choose $\beta : E^\bullet \to L^\bullet$ with $E^\bullet$ a bounded complex of finite free modules such that $H^i(\beta)$ is an isomorphism for $i > m$ and surjective for $i = m$. By Derived Categories, Lemma 064E we can find a map $\gamma : P^\bullet \to E^\bullet$ such that the diagram

$$\begin{array}{ccc} K^\bullet & \longrightarrow & L^\bullet \\ \uparrow & & \uparrow{\scriptstyle \beta} \\ P^\bullet & \xrightarrow{\gamma} & E^\bullet \end{array}$$

is commutative in $D(R)$. The cone $C(\gamma)^\bullet$ is a bounded complex of finite free R-modules, and the commutativity of the diagram implies that there exists a morphism of distinguished triangles

$$(P^\bullet, E^\bullet, C(\gamma)^\bullet) \longrightarrow (K^\bullet, L^\bullet, M^\bullet).$$

It follows from the induced map on long exact cohomology sequences and Homology, Lemmas 05QA and 05QB that $C(\gamma)^\bullet \to M^\bullet$ induces an isomorphism on cohomology in degrees $> m$ and a surjection in degree m. Hence $M^\bullet$ is m-pseudo-coherent.

Assertions (2) and (3) follow from (1) by rotating the distinguished triangle. □